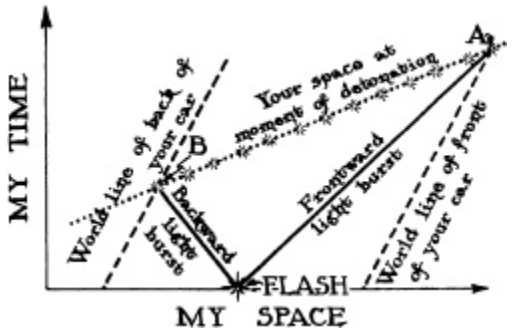


Note that the locations of the detonations (your space at a specific moment of your time) are the same in the above spacetime diagrams as in Figure 1.3. This justifies the asserted mixing of space and time discussed in the text.

How can it possibly be that I see your time flow slowed, while you see mine slowed? How is that logically possible? And how can I see your space contracted, while you see my space contracted? The answer lies in the relativity of simultaneity. You and I disagree about whether events at different locations in our respective spaces are simultaneous, and this disagreement turns out to mesh with our disagreements over the flow of time and the contraction of space in just such a way as to keep everything logically consistent. To demonstrate this consistency, however, would take more pages than I wish to spend, so I refer you, for a proof, to Chapter 3 of Taylor and Wheeler (1992).

How is it that we as humans have never noticed this weird behavior of space and time in our everyday lives? The answer lies in our slowness. We always move relative to each other with speeds far smaller than that of light (299,792 kilometers per second). If your car zooms down Colorado Boulevard at 150 kilometers per hour, I should see your time flow dilated and your space contracted by roughly one part in a hundred trillion (1×10^{-14})—far too little for us to notice. By contrast, if your car were to move past me at 87 percent the speed of light, then (using instruments that respond very quickly) I should see your time flow twice as slowly as mine, while you see my time flow twice as slowly as yours; similarly, I should see everything in your car half as long, east–west, as normal, and you should see everything in my police station half as long, east–west, as normal. Indeed, a wide variety of experiments in the late twentieth century have verified that space and time do behave in just this way.